Exp 1 Content analysis

Analyze the content of social media data to determine what topics are being given data set using topic modelling, keyword extractor

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from wordcloud import WordCloud
import matplotlib.pyplot as plt
df = pd.read_csv("C:/Users/PRIYANKA/Desktop/sma_lab/SMA_2_data.csv")
print(df.info())
# extract 10 keywords and removes common english words like a, is, the
vect = TfidfVectorizer(max_features=10, stop_words='english')
x=vect.fit_transform(df['Tweet_Text'])
keywords = vect.get_feature_names_out()
print("Top keywords: ", keywords)
# Create a WordCloud from the content of the posts
wordcloud = WordCloud(width=800, height=400,
background_color='white').generate(' '.join(df['Tweet_Text']))
# Display the WordCloud
plt.figure(figsize=(10, 6))
plt.imshow(wordcloud, interpolation='bilinear')
plt.axis('off')
plt.show()
keyword_counts = x.sum(axis=0).A1  # Sum of word frequencies across all
documents
plt.bar(keywords,keyword_counts)
plt.show()
```

Exp 2 Location Analysis

Analyze the location data associated with tweets to understand where particular location are most prevalent.

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("C:/Users/PRIYANKA/Desktop/sma_lab/SMA_LOC_DATA.csv")
print(df.info())
# count frequency of each location
count = df['Location'].value_counts()
print(count)
# plot top locations
count.plot(kind='bar')
plt.show()
```

Exp 3 Trend Analysis

Analyze the time data associated with social media and analyse its trends.

```
import pandas as pd
import matplotlib.pyplot as plt
df =
pd.read_csv('C:/Users/PRIYANKA/Desktop/sma_lab/cleaned_youtube_data.csv')
print(df.info())
# convert date to datetime format
df['date'] = pd.to_datetime(df['date'])
# extract year, month and day
df['year'] = df['date'].dt.year
df['month'] = df['date'].dt.month
df['day'] = df['date'].dt.day
# trend of average likes over time used mean for avg, use sum for total
like_count = df.groupby(['year', 'month'])['like_count'].mean()
like_count.plot(kind='line')
plt.show()
```

Exp 4 Hashtag popularity analysis

Determine which hashtags are most popular among different user groups

Exp 5 Sentiment Analysis for given dataset (negative, positive)

```
from nltk.sentiment.vader import SentimentIntensityAnalyzer
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('C:/Users/PRIYANKA/Desktop/sma_lab/cleaned_comments1.csv')
print(df.info())
df = df.dropna(subset=['processed_text']).reset_index(drop=True)
sa = SentimentIntensityAnalyzer()
def classify_sentiment(text):
   score = sa.polarity_scores(str(text))['compound']
   if score >= 0.05:
       return 'positive'
   elif score <= -0.05:
       return 'negative'
   else:
       return 'neutral'
df['sentiment'] = df['processed_text'].apply(classify_sentiment)
sns.countplot(df, x=df['sentiment'], order=['positive', 'neutral',
'negative'])
plt.show()
```

Exp 6 User engagement analysis

Analyze how users engage with content on social media to understand what types of content are most engaging.

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv('C:/Users/PRIYANKA/Desktop/sma_lab/Tweets.csv')
print(df.info())
# print(df['tweet_created'].head())
df['tweet_created'] = pd.to_datetime(df['tweet_created'])
df['hour'] = df['tweet_created'].dt.hour
# average engagement of retweets per hour
retweet_count_by_hour = df.groupby('hour')['retweet_count'].mean()
retweet_count_by_hour.plot(kind='bar')
plt.show()
# which user retweeted most
user_retweet = df.groupby('name')['retweet_count'].sum().head(20)
user_retweet.plot(kind='bar')
plt.show()
```

Exp 7 Exploratory Data Analysis

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv('C:/Users/PRIYANKA/Desktop/sma_lab/Tweets.csv')
from nltk.sentiment.vader import SentimentIntensityAnalyzer
sa = SentimentIntensityAnalyzer()


def sentiment_count(text):
   score = sa.polarity_scores(str(text))['compound']
   if score >= 0.05:
       return 'positive'
   elif score <= -0.05:
       return 'negative'
   else:
       return 'neutral'


df['score'] = df['text'].apply(sentiment_count)

# overall sentiment analysis
sentiment_pie = df['airline_sentiment'].value_counts()
# plt.pie(sentiment_pie, labels=sentiment_pie.index)
# plt.show()

# sentiment analysis of text by airline
# sns.boxplot(df, x=df['airline'], y=df['score'])
# plt.show()

# tweeting pattern
```

```
df['tweet_created'] = pd.to_datetime(df['tweet_created'])
df['hour'] = df['tweet_created'].dt.hour
# sns.countplot(df, x=df['hour'])
# plt.show()

# airline count
airline_count = df['airline'].value_counts()
# sns.countplot(df,x=df['airline'])
# plt.show()
```

Exp 8 Brand analysis

analyze the conversation around a particular brand for given dataset

```
import pandas as pd
from nltk.sentiment.vader import SentimentIntensityAnalyzer
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv('C:/Users/PRIYANKA/Desktop/sma_lab/Tweets.csv')
print(df.info())
sa = SentimentIntensityAnalyzer()


def sentiment(text):
   score = sa.polarity_scores(str(text))['compound']
   if score >= 0.05:
       return 'positive'
   elif score <= 0.05:
       return 'negative'
   else:
       return 'neutral'


df['sentiment'] = df['text'].apply(sentiment)
# brand analysis according to sentiment per airline
sns.countplot(df, x='airline', hue='sentiment')
plt.show()
```

Exp 9 Competitor Analysis

Exp 10 Community detection/ influential analytics